



Habib University - City Campus
Course: CS/CE 355/373: Database Systems
Examination: Midterm Exam – Fall 2024
Exam Date: Wednesday, October 9th, 2024
Exam Time: 5:45 PM – 7:00 PM

Total Marks: 100 Marks

Duration: 75 Minutes

Name: _____ Student ID: _____ Section: _____

DO NOT TURN OVER UNTIL INSTRUCTED.

Please read the following instructions carefully.

1. Keep track of the time.
2. Use of mobile devices is strictly prohibited.
3. **Please do not use a pencil or a red pen.**
4. Place your ID card on your desk in front of you.
5. You may keep writing material and a snack with you on your desk.
6. Please submit your devices in your bag at the front of the examination room.
7. Acquisition of answers through unfair means will automatically cancel your exam.
8. This sheet contains 4 questions for 60 points on 5 printed sides including this one.

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Question 1:

Create an ER Diagram for the following scenario. Identify entities and their relationships with appropriate attributes and keys.

You have been hired by a media company to build software for their advertisement platform. The company plans to partner with 5,000 businesses to place LED TVs on their premises. Depending on the size of the premises, each business might have one or more LED screens. Some businesses may have multiple locations, and the company wants to track the number of LEDs at each premise.

The media company will publish 30-second video advertisements. When creating an advertisement, the required information includes a unique name, a short description, and the video file itself. The system must also store details about which admin user created the ad, the date it was created, the last update date, and the associated brand. Each brand can have multiple ads.

Advertisements will be categorized into categories and subcategories. A subcategory can only belong to one category, but a category can have multiple subcategories.

Question 2:

Consider the following relational schema about international cricket matches and players:

Countries (Country, Continent, Capital)

Players (PlayerID, Name, Country (FK to Countries), BattingAverage, BowlingAverage)

MatchRecords (MatchID, PlayerID (FK to Players), RunsScored, WicketsTaken, Stadium)

Assume all the international cricketers have a unique player ID. Also, assume that a cricket player cannot represent two different countries.

Specify the following queries in relational algebra:

- List the name, country, and batting average of the player with ID: 804.
- List all the stadiums where the player with ID 804 has scored a century.
- List the ID and names of all the players from Pakistan whose batting average is less than 8.04.
- List the ID and names of all the players from Asia whose batting average is less than 8.04.
- List all the stadiums where the player with ID 804 has scored a century but has never taken a wicket.

Question 3:

Imagine you are employed as a Data Analyst at Habib University responsible for managing students' extracurricular activities information. A student can be a part of multiple extracurricular activities. Each of these activities is scheduled at different timings. During each time slot, there is always an Activity Lead who overlooks all the participating students in that group. A student can attend any of the time slots, as per their availability, so no specific slot will always have the same set of students. The university administration is using a table to store this information. Some of the rows from that table are given in Table 1 for reference (**Table is given at last page of this document**).

Analyze the provided extracurricular activities data table, and derive a relation schema for it in 1NF, by specifying the primary keys. Give reasons for choosing the primary keys. Identify all relevant functional dependencies to justify your decomposition. If there are no dependencies, please state that explicitly.

Normalize the schema up to the Third Normal Form (3NF) by creating new tables. Provide schemas for 2NF, and 3NF.

Question 4:

Answer all MCQs given below.

1. While translating an ER Diagram to schema, where should the foreign key be placed when resolving a one-to-many relationship?
 - (a) On the "one" side of the relationship
 - (b) On the "many" side of the relationship
 - (c) In both entities
 - (d) There should be no foreign key
2. Which normal form removes any partial dependencies on the primary key?
 - (a) First Normal Form (1NF)
 - (b) Second Normal Form (2NF)
 - (c) Third Normal Form (3NF)
 - (d) Boyce-Codd Normal Form (BCNF)
3. Which of the following is/are true about a foreign key?
 - (a) A foreign key always has unique values in a table.
 - (b) A foreign key must reference a primary key in another table.
 - (c) A foreign key is used to define the uniqueness of rows in the table.
 - (d) A foreign key can reference any non-key attribute in the other table.
4. In a relational database, which of the following statements about candidate keys is correct?
 - (a) All candidate keys are super keys.
 - (b) All super keys are candidate keys.
 - (c) A candidate key can only consist of one attribute.
 - (d) There can only be one candidate key per table.
5. A university has multiple departments, and each department can have multiple professors. However, each professor belongs to only one department. What type of relationship exists from the Department to Professor entities?
 - (a) One-to-One
 - (b) One-to-Many
 - (c) Many-to-One
 - (d) Many-to-Many
6. In a school's database, each student can enroll in multiple courses, and each course can have multiple students enrolled. What type of relationship is represented from the Student to Course entities?
 - (a) One-to-One
 - (b) One-to-Many
 - (c) Many-to-One
 - (d) Many-to-Many
7. What is the result of applying the Cartesian Product ($\times$) between two relations?
 - (a) A set of tuples that appear in both relations.
 - (b) A relation that contains only matching tuples from both relations.
 - (c) A relation containing all possible combinations of tuples from both relations.

- (d) A set of tuples that appear in the first relation but not in the second.
8. Which of the following is an example of a transitive dependency in a relation?
- (a) $A \rightarrow B$ and $B \rightarrow C$, therefore $B \rightarrow C$
 - (b) $A \rightarrow C$ and $C \rightarrow B$, therefore $A \rightarrow B$
 - (c) $A \rightarrow B$, and $B \rightarrow C$, but $A \rightarrow C$ does not hold
 - (d) $A \rightarrow B$, and $B \rightarrow C$, so no further dependencies exist
9. In a Library database, the relation Book has the following attributes: BookID, ISBN, Title, and Author. Consider the following set of superkeys: $\{\text{BookID}, \text{ISBN}\}$, $\{\text{BookID}, \text{Title}\}$, $\{\text{ISBN}, \text{Author}\}$, $\{\text{BookID}\}$. Which of the following sets of attributes can be considered a candidate key?
- (a) $\{\text{BookID}\}$
 - (b) $\{\text{ISBN}\}$
 - (c) $\{\text{BookID}, \text{ISBN}\}$
 - (d) $\{\text{ISBN}, \text{Author}\}$
10. We have two relation Student (with 12 tuples/rows) and Course (with 4 tuples/rows). What will be the number of tuples/rows in the result of Cartesian Product of $\text{Student} \times \text{Course}$?
- (a) 3
 - (b) 4
 - (c) 12
 - (d) 48

StudentID (PK)	StudentName	DOB	Year	Activity	ActivityType	ActivityTimings	ActivityLead	ActivityJoiningDate	ActivityLevel	GuardianID	GuardianName	GuardianContact
101	Ammara Javed	13-02-2005	1st	Basketball	Sports	Morning	Mr. X	31-08-2024	Intermediate	1	Javed Jaffrey	555-1234
						Afternoon	Mr. Y					
				Drama Club	Performance	Evening	Ms. A	15-09-2024	Beginner			
102	Bilal Syed	22-08-2004	2nd	Chess Club	Hobby	Morning	Mr. B	10-09-2023	Advanced	2	Mariam Syed	555-5678
				Coding	Educational	Afternoon	Mr. C	01-01-2024	Intermediate			
						Evening	Ms. D					
				Debate	Performance	Afternoon	Ms. G	15-06-2024	Beginner			
103	Dawood Waqar	10-09-2005	1st	Football	Sports	Morning	Mr. E	10-08-2024	Advanced	3	Sarah Bilal	555-4321
						Afternoon	Mr. F					
				Robotics	Educational	Morning	Mr. H	01-09-2024	Beginner			
						Evening	Mr. I					
104	Erum Pervez	20-01-2002	4th	Drama Club	Performance	Evening	Ms. A	01-09-2021	Advanced	4	Pervez Waheed	555-9876
				Cooking Club	Hobby	Afternoon	Ms. J	19-01-2024	Beginner			
105	Faheem Haq	23-05-2003	3rd	Debate	Performance	Afternoon	Ms. G	04-12-2023	Beginner	5	Nadeem Haq	555-6543
				Basketball	Sports	Morning	Mr. X	22-11-2022	Advanced			
						Afternoon	Mr. Y					
				Coding	Educational	Afternoon	Mr. C	07-10-2022	Advanced			
						Evening	Ms. D					
106	Ghazal Khan	24-07-2004	2nd	Chess Club	Hobby	Morning	Mr. B	26-10-2023	Intermediate	6	Mikaal Khan	555-3456
				Robotics	Educational	Morning	Mr. H	19-06-2024	Beginner			
						Evening	Mr. I					
				Cooking Club	Hobby	Afternoon	Ms. J	29-04-2024	Beginner			
107	Harish Chandra	01-01-2022	4th	Football	Sports	Morning	Mr. E	18-03-2021	Advanced	7	Laxman Chandra	555-7890
						Afternoon	Mr. F					
108	Ibrahim Shah	21-04-2003	3rd	Basketball	Sports	Morning	Mr. X	12-12-2022	Intermediate	8	Adam Shah	555-6789
						Afternoon	Mr. Y					
109	Javeria Merchant	20-09-2004	2nd	Cooking Club	Hobby	Afternoon	Ms. J	17-11-2023	Intermediate	9	Ehtasham Merchant	555-1239
110	Kiran Zahid	29-03-2005	1st	Coding	Educational	Afternoon	Mr. C	30-08-2024	Beginner	10	Zahid Siddiqui	555-0123
						Evening	Ms. D					
				Drama Club	Performance	Evening	Ms. A	30-08-2024	Beginner			
111	Monis Shahmeer	16-04-2003	3rd	Coding	Educational	Afternoon	Mr. C	15-06-2023	Intermediate	11	Shahmeer Zulfiqar	555-2345
						Evening	Ms. D					
				Football	Sports	Morning	Mr. E	18-12-2022	Advanced			
						Afternoon	Mr. F					
112	Abdul Ahad	13-05-2002	4th	Football	Sports	Morning	Mr. E	25-08-2021	Advanced	12	Abdul Samad	555-4567
						Afternoon	Mr. F					
				Basketball	Sports	Morning	Mr. X	27-08-2021	Advanced			
						Afternoon	Mr. Y					
				Debate	Performance	Afternoon	Ms. G	01-01-2022	Intermediate			